

Pattern making using Looping and function.

Print name in all possible combinations of pattern using function

```
def akkrat(name):  
    for i in range(len(name)):   
        for j in range(0,i+1):  
            print(name[i],end='')  
        print()  
akkrat('akshara')
```

```
a  
kk  
sss  
hhhh  
aaaaa  
rrrrrr  
aaaaaaa
```

```
def aklat(name):  
    for i in range(len(name)):   
        for k in range(len(name),i,-1):  
            print(' ',end='')  
        for j in range(i+1):  
            print(name[i],end='')  
        print()  
aklat('akshara')
```

```
      a  
     kk  
    sss  
   hhhh  
  aaaaa  
 rrrrrr  
aaaaaaa
```

```
[1]: def iakkrat(name):  
      for i in range(len(name)-1,-1,-1):  
          for j in range(0,i+1):  
              print(name[i],end='')  
          print()  
      iakkrat('akshara')
```

```
aaaaaaa  
rrrrrrr  
aaaaa  
hhhhh  
sss  
kk  
a
```

```
[1]: def iaklat(name):  
      for i in range(len(name)-1,-1,-1):  
          for k in range(7,i,-1):  
              print(' ',end='')  
          for j in range(0,i+1):  
              print(name[i],end='')  
          print()  
      iaklat('akshara')
```

```
aaaaaaa  
rrrrrrr  
aaaaa  
hhhhh  
sss  
kk  
a
```

```
def akratupper(name):  
    for i in range(len(name)):   
        for j in range(0,i+1):  
            print(name[i].upper(),end='')  
        print()  
akratupper('akshara')
```

```
A  
KK  
SSS  
HHHH  
AAAAA  
RRRRRR  
AAAAAAA
```

```
def aklatupper(name):  
    for i in range(len(name)):   
        for k in range(len(name),i,-1):  
            print(' ',end='')  
        for j in range(i+1):  
            print(name[i].upper(),end='')  
        print()  
aklatupper('akshara')
```

```
      A  
     KK  
    SSS  
   HHHH  
  AAAAA  
 rrrrrr  
AAAAAAA
```

```
def iakratupper(name):
    for i in range(len(name)-1,-1,-1):
        for j in range(0,i+1):
            print(name[i].upper(),end='')
        print()
iakratupper('akshara')
```

```
AAAAAAA
RRRRRRR
AAAAA
HHHH
SSS
KK
A
```

```
def iakratupper(name):
    for i in range(len(name)-1,-1,-1): #-1 can be used as a stop value
        for j in range(len(name),i,-1):# as space is always in between
            print(' ',end='')
        for j in range(0,i+1): #no of columns is i+1
            print(name[i].upper(),end='') #name[i] is to print the character
        print()
iakratupper('akshara')
```

```
AAAAAAA
RRRRRRR
AAAAA
HHHH
SSS
KK
A
```

```
def akpyramidupper(name):
    for i in range(len(name)):
        for k in range(len(name),i,-1):
            print(' ',end=' ')
        for j in range(i+1):
            print(name[i].upper(),end=' ')
        print()
akpyramidupper('akshara')
```

```
      A
     K K
    S S S
   H H H H
  A A A A A
 R R R R R R
A A A A A A A
```

```
def akpyramid(name):
    for i in range(len(name)):
        for k in range(len(name),i,-1):
            print(' ',end=' ')
        for j in range(i+1):
            print(name[i],end=' ')
        print()
akpyramid('akshara')
```

```
      a
     k k
    s s s
   h h h h
  a a a a a
 r r r r r r
a a a a a a a
```

The factor of a given 2-digit number

```
def factor():
    number= int(input('type the 2 digit number to find the factor: '))
    for i in range(1,number+1):
        if number%i==0:
            print(i)
factor()
```

```
type the 2 digit number to find the factor: 23
1
23
```

```
factor()
```

```
type the 2 digit number to find the factor: 52
1
2
4
13
26
52
```

Using lambda function to find the biggest of four numbers.

```
big=lambda a,b,c,d: a if a>b and a>c and a>d else b if b>c and b>d else c if c>d else d  
big(20,30,40,50)
```

50

```
big(112,36,20,200)
```

200

Cube and Cube root

```
num=[1,2,3,4,5]  
cube=list(map(lambda a:a**3,num))  
cube
```

[1, 8, 27, 64, 125]

```
num=[1, 8, 27, 64, 125]  
cuberoot=list(map(lambda a:a**(1/3),(num)))  
cuberoot
```

[1.0, 2.0, 3.0, 3.9999999999999996, 4.999999999999999]